

Educational Ambition, Marital Sorting, and Inequality*

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Abstract

This paper revisits the link between education-based marriage market sorting and income inequality. Leveraging Danish administrative data, we develop a novel categorization of “ambition types” that is based on the labor market returns of detailed educational programs. In contrast to commonly-used education-based categorizations, we find a substantial increase in positive assortative matching by educational ambition over time. Moreover, changes in household composition by ambition explain more than 35% of increasing inequality since 1980. We conclude that the mapping from education to types matters crucially for conclusions about how education-based marriage market sorting contributes to rising income inequality.

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1 Introduction

An ongoing debate questions the contribution of education-based positive assortative matching (PAM) in the marriage market to rising household income inequality. Some studies find evidence that assortative matching on education has strengthened in recent decades and argue that this has contributed to rising income inequality across households (Fernández and Rogerson, 2001; Greenwood et al., 2014, 2016; Mare, 2016; Chiappori et al., 2017; Hryshko et al., 2017; Ciscato and Weber, 2020; Calvo et al., 2024). Other papers argue against both findings (Kremer, 1997; Breen and Salazar, 2011; Breen and Andersen, 2012; Eika et al., 2019; Gihleb and Lang, 2020). We contribute to this debate by showing that *how* educational data are used to capture relevant traits for marriage market matching influences conclusions about the interplay of marital sorting and inequality.

We leverage rich administrative data from Denmark to go beyond the usual educational categories based on either levels (e.g., college) or fields of study (e.g., Social Sciences). Motivated by models of household behavior in which families value education due to its earnings potential, we develop a novel categorization that captures differences in the *labor market returns* to human capital across educational *programs* (e.g., Economics). Specifically, we group more than 1800 educational programs into four categories that exhibit similar *wage dynamics*, as summarized by the average starting wages, w_0 , and average wage growth, g , of graduates. These measures map directly into the labor market payoffs of each program: w_0 captures the price of human capital, and wage growth g reflects the returns to career-shaping choices. Thus, w_0 and g capture early-career returns to experience and, in this sense, serve as sufficient statistics for the labor-market value of graduating from a specific educational program. We interpret differences in this value across programs as reflective of graduates' potential for labor market success and *career ambition*.

Since household resources are determined by *returns* to human capital that differ across programs, it is natural to ask whether household income inequality has been affected by changes in the composition of households based on these very returns. Using our categorization, we find that PAM increased substantially between 1980–2018 (by 20%), and changes in the overall composition of households account for 35% of the increase in income inequality over this period. In contrast, PAM by educational levels or fields exhibits no trend since the mid-1990s and the changing composition of households based on these categorizations can account only for a small fraction of inequality growth. Alternative classifications based on work hours or lifetime earnings show no trend in PAM either, which suggests that spouses increasingly sort on labor

market returns per work hour rather than work schedules or lifetime income.

Methodologically, we build on the literature that emphasizes heterogeneity in labor market outcomes across education programs (Altonji et al., 2014, 2016) and exploit detailed data on labor market outcomes of graduates at the level of the educational program. Programs are defined based on ISCED codes, which identify more than 1800 unique education programs in Denmark. Examples include the vocational training of *carpenters*, professional degrees held by *nurses*, and 5-year university degrees in *law* and *business*. We define four types by grouping programs based on similarity of wage dynamics for typical graduates—average starting wages and wage growth—using k-means clustering, a well-established and popular partitioning method in machine learning and computer science (Steinley, 2006). This method has recently been introduced to economic research (Bonhomme and Manresa, 2015) and applied to categorize worker and firm types in the labor market (Bonhomme et al., 2019, 2022). To our knowledge, we are the first to apply this method to construct education-based types to study marriage market matching. Our method allows us to collapse multiple wage moments into one type and thereby simplify the analysis of assortative matching.¹

In the first part of our analysis, we show that our method successfully clusters education programs into four clearly distinct groups based on whether starting wages and wage growth are high or low. We label our categorization *educational ambition* because we interpret the four groups as capturing differences in the potential for labor market success, which we also confirm in the data. In contrast, groups based on educational *levels* or *fields* mask significant heterogeneity in starting wages and wage growth.

In the second part of the analysis, we compare trends in assortative matching across categorizations and study how the composition of households contributes to the rise in between-household inequality. To this end, we consider the role of *trends* along three margins that influence marital sorting as reflected by *the composition of households*: the marginal distributions of education-based types by gender, the share of singles by gender and education (the *extensive margin*, i.e., who marries), and the frequency of couples by spouses' joint education types (the *intensive margin*, i.e., who marries whom).

Our first main finding is that the degree of assortative matching has increased significantly based on educational ambition, but has remained stable based on educational levels or fields. Thus, conclusions about assortative matching trends crucially depend on the categorization. To study these trends in PAM and compare them across categorizations, we follow Eika et al.

¹Multidimensional sorting in labor and marriage markets has been explored, for example, by Lindenlaub (2017), Ciscato and Weber (2020), Low (2024), Foerster et al. (2024), and Chiappori et al. (2024).

(2019), [Almar and Schulz \(2024\)](#), and [Chiappori et al. \(2025\)](#). We measure the degree of PAM by the weighted sum of the matching frequencies of same-type couples relative to random matching (likelihood ratios).

Our second main finding is that changes in the three marriage market margins based on educational ambition contribute substantially more to the overall increase in income inequality between 1980 and 2018 in Denmark than changes in household composition based on educational level or field. Methodologically, we follow [DiNardo et al. \(1996\)](#) and [Fortin et al. \(2011\)](#) and compare observed between-household inequality measures by year to counterfactual measures that we calculate by re-weighting households such that their composition stays fixed as in 1980. Note that the degree of assortative matching is also held fixed in this counterfactual. We further assess the separate roles of trends in the marginal type distributions and marital choices to explain inequality. To this end, we follow [Chiappori et al. \(2020a\)](#) and combine our counterfactual re-weighting method with a structural marriage market matching model of the Separable Extreme Value (SEV) type ([Choo and Siow, 2006](#); [Chiappori et al., 2017, 2020b, 2022](#)), which allows us to decompose changes in the composition of households into changes in the value of marriage relative to remaining single (marital surplus) and changes in the marginal type distributions. We find that the evolution of marital surplus contributes equally to the rise in inequality, regardless of how we construct types. In contrast, changes in the marginal distributions mitigate inequality through the lens of education levels or fields—but not based on educational ambition.

We complement our analysis by considering two alternative ways to measure the labor market returns of educational programs. First, we consider the similarity of graduates' typical working hours and their irregularity. Second, we group programs based on average lifetime earnings. We show that these classifications deviate substantially from our ambition baseline because they fail to capture differences in the returns to work hours and across career paths that generate similar lifetime earnings. Consequently, these classifications understate the increase in positive assortative matching relative to our benchmark.

Our findings are also robust to a variety of alternative ways of constructing ambition types. In particular, we construct gender- and cohort-specific ambition types, vary the numbers of types, and use alternative clustering variables, e.g., part-time penalties ([Goldin, 2014](#)) or manager premia. Our approach is also robust to using more aggregated units of observation at educational levels by sub-fields, which showcases the broader applicability of our method.

We contribute to the literature on the value of college degrees in the marriage market ([Kirkeboen et al., 2022, 2016](#); [Artmann et al., 2021](#); [Nielsen and Svarer, 2009](#); [Wiswall and Zafar,](#)

2021; Seiver and Sullivan, 2020; Han and Qian, 2022). Unlike these papers, our measure of educational ambition considers programs from all levels of education—from compulsory schooling to graduate school—allowing us to study trends in the composition of households for the broad population. Furthermore, we build on the literature that underscores the value of (permanent) income for assortative matching (e.g., Chiappori et al., 2022) and contribute a measure that differentiates between career paths.

Moreover, we contribute to the debate on the relationship between trends in marriage market sorting and inequality by showing that the choice of how to map data on educational attainment into types affects the conclusions. All three categorizations of education-based types that we consider—ambition, levels, and fields—use information from the most advanced program that individuals graduated from. Still, conclusions about how the composition of households has changed and influenced inequality differ significantly. First, the literature that considers the level of education (Breen and Salazar, 2011; Breen and Andersen, 2012; Greenwood et al., 2014; Eika et al., 2019; Chiappori et al., 2020a) and the field of study (Seiver and Sullivan, 2020; Artmann et al., 2021; Han and Qian, 2022) yields mixed results. In contrast, our novel ambition types reveal an upward trend in positive assortative matching and a significant contribution of changing household composition to the increment in inequality. Second, we build on Burtless (1999) and Chiappori et al. (2020a) and include trends in single-headed households into our analysis of the link between family composition and income inequality. We extend their work by showing that trends in single shares by education and gender differ depending on how education-based types are defined, and this partly explains the differences in conclusions about the link between household composition and inequality across categorizations.

Finally, our insight that the categorization of types matters to capture the relationship between marriage market sorting and inequality can inform the literature on the link between assortative matching and inter-generational mobility (Bailey and Lin, 2022; Binder et al., 2022; Gayle et al., 2015). Because these studies compare groups defined by, for example, race or income, the definition of the trait on which people sort in the marriage market is potentially important for their conclusions as well.

2 Data

We use register data for the entire population of Denmark in the period 1980–2018. The data are a yearly panel and unique person identifiers allow us to combine information on individuals’ education, marital status, fertility, labor market outcomes, and the identity and

characteristics of their marriage or cohabiting partner (BEF, [Statistics Denmark, 1980–2018](#)). We add information on work hours from the Danish Labor Force Survey (LFS) ([LFS, Statistics Denmark, 2000–2018](#)) for a subset of workers. We next describe the key variables we construct for our analysis and provide further details in Online Appendix A.

The sample includes all residents aged 19–60 from 1980 to 2018, with on average about 3 million individuals per year, of whom around 1.8 million are individuals in opposite-sex couples (married or cohabiting). In this sample, we measure individual education as the most advanced completed *educational program*, defined by four-digit ISCED codes (about 1,800 programs in Denmark, [UDDA, Statistics Denmark \(1980–2018\)](#)).² The most popular programs include vocational degrees such as *bank advisor*, *carpenter* or *office clerk*, bachelor’s degrees like *nurse* or *pre-school teacher*, and master’s degrees in *business*, *law* or *medicine*.³

For our main categorization, we use program graduates’ *hourly wages* from the employment register ([IDAN, Statistics Denmark, 1980–2018](#)) and compute program-level wage outcomes after residualizing log wages on year dummies (with base year 2000) to remove time effects.⁴ The representative LFS (2000–2018)⁵ provides exact work hours and indicators for evening, weekend, and overtime work, which we use for an alternative hours-based categorization.

We analyze income inequality at the household level, considering both singles and opposite-sex couples. Our measure of household income is yearly labor income from both regular employment and self-employment according to the income register ([IND, Statistics Denmark, 1980–2018](#)). For couples, we use the sum of both spouses’ income and apply the OECD equivalence scale to make one-person and two-person households comparable.

3 Ambition Types

Previous research has established that education is a valuable trait in the marriage market. It predicts both earnings potential and future time investments into career and family ([Chiappori et al., 2018](#); [Gayle and Shephard, 2019](#); [Calvo, 2023](#); [Calvo et al., 2024](#); [Reynoso, 2024](#)). In this section, we construct a wage-based categorization of educational programs that reflects heterogeneity in life-cycle returns to human capital and the prospects for labor market success

² We exclude 2.6% of individuals with missing education information.

³ Some programs are tied to one particular university and thus location, but most programs are spread around the country. We distinguish university degrees by institutions in an extension; see Online Appendix B.5.

⁴ Hourly wages are calculated based on administrative information on labor income and hours worked in the employment register. Hours worked reflect contractual hours, i.e., part-time work is captured but not overtime.

⁵ Approximately 72,000 participants are surveyed on an annual basis. The sample is weighted to ensure that it is representative of the entire population of Denmark.

across programs. Importantly, different wage dynamics across programs allow us to distinguish the returns of different career trajectories. In turn, these returns capture the opportunity cost of time at home. In contrast, common definitions of education based on educational level or field of study mask substantial heterogeneity in wage dynamics, which we find to be crucial for understanding trends in assortative matching and inequality.

3.1 Conceptual Framework

To construct education-based types, suppose that men and women are distinguished by their education program, $i \in \mathcal{P} = \{Program_1, Program_2, \dots, Program_I\}$. I is the total number of programs. Each program i is characterized by an N -dimensional vector of observable characteristics, $x_i = (x_{1i}, x_{2i}, \dots, x_{Ni})$, and a dataset at the program level is defined as $x : \{x_i\}_{i \in \mathcal{P}}$. Examples of such characteristics include the length of the educational program, the field of study, the level of education, and the labor market outcomes of graduates.

For both tractability and ease of interpretation, education programs are commonly grouped into a small number of types based on their similarity in *selected* characteristics, $\tilde{x} \subset x$. Formally, let $\mathcal{T}_{\tilde{x}} : \tilde{x} \rightarrow t = \{Type_1, Type_2, \dots, Type_T\}$ be a mapping that defines $T \ll I$ types by grouping the I programs based on their similarity in the sub-vector of observable characteristics $\tilde{x}$.

The literature commonly uses the mapping $\mathcal{T}_{Levels}$, which maps programs based on similarity in one characteristic, namely, their level of education (see, e.g., [Fernández and Rogerson, 2001](#); [Greenwood et al., 2014, 2016](#); [Eika et al., 2019](#)). One way to implement this mapping is to choose four types, $t^{Levels} = \{Primary, Secondary, Bachelor, Master\&PhD\}$. Another popular one-dimensional mapping is $\mathcal{T}_{Fields}$ (see, e.g., [Seiver and Sullivan, 2020](#); [Artmann et al., 2021](#); [Kirkeboen et al., 2022](#); [Han and Qian, 2022](#)). It focuses on post-secondary education and groups programs by field of study, mapping the narrow fields of individual programs, $\tilde{x}_i = field_i$, into broader groups of fields (e.g., STEM, health, humanities, social sciences, business). This yields the mapping $t^{Fields} = \{Field_1, Field_2, \dots, Field_T\}$.

3.2 Construction of Ambition Types

We take advantage of our rich data and construct the ambition types using average labor market outcomes of graduates at the program level. We use the *average starting wage*—denoted by w_0 —and *average wage growth over the early career*—denoted by g .

Formally, for each of the more than 1800 programs of education i we observe $\tilde{x}_i = (w_{0,i}, g_i)$,

calculated using information on all individuals who completed their education after 1980. As explained in more detail in Online Appendix A.2, we first residualize log hourly wages and then compute w_0 as the average hourly wage of program graduates during the first five years in the labor force.⁶ To calculate average wage growth, g , we measure the percentage change between w_0 and w_1 , where w_1 is the average hourly wage of program graduates in years 9-11 in the labor force.⁷ We average across years for both w_0 and w_1 to smooth out year-to-year variation that is unrelated to worker productivity.

In our benchmark analysis, we construct four types using the mapping $\mathcal{T}(w_0, g)$. To implement the mapping, we cluster programs based on standardized starting wages and wage growth using the k-means algorithm (Steinley, 2006). This method minimizes the within-cluster variation in the two dimensions and thus produces homogeneous groups in terms of starting wages and wage growth. We denote this mapping $\mathcal{T}_{Ambition}$, which includes the following four types:

$$t^{Ambition} = \{(low\ w_0, low\ g), (high\ w_0, low\ g), (low\ w_0, high\ g), (high\ w_0, high\ g)\}.$$

Panel (a) in Figure 1 plots the mapping $\mathcal{T}_{Ambition}$. It shows how the program-specific (w_0, g) tuples map into our ambition types. The plot locates approximately 1800 programs in the space of standardized starting wages (horizontal axis) and standardized wage growth (vertical axis). We distinguish the resulting four types with different colors and markers. The mapping delivers groups that are clearly (and by construction) distinguished by whether starting wages and wage growth are low or high. Our interpretation is that graduates from high-starting-wages/high-wage-growth programs signal *career-ambition* in the marriage market by their choice of educational program. Thus, we label our categorization *educational ambition*.

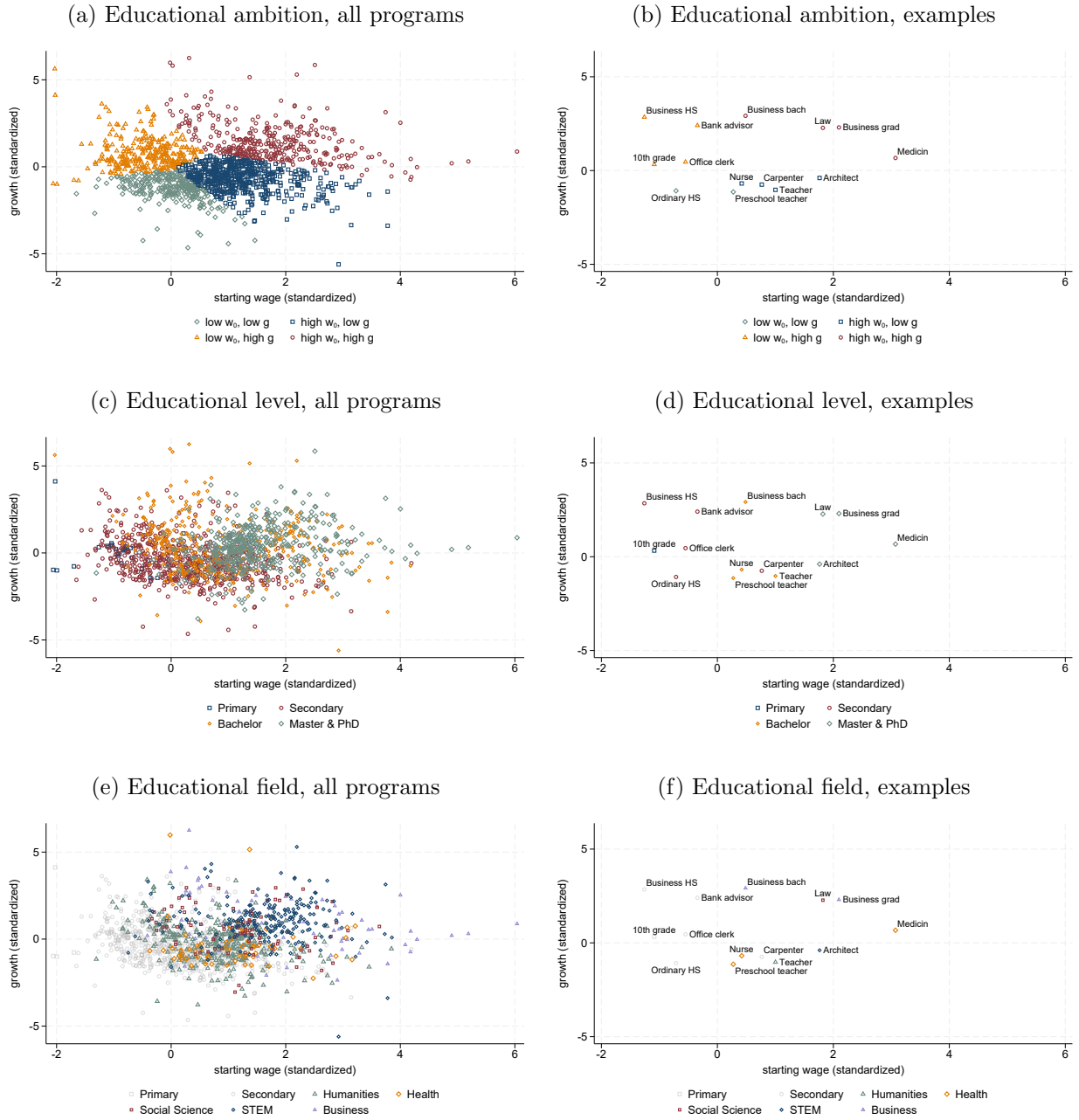
Online Appendix Table A.1 describes the four educational ambition types by population shares, gender, income, and parental wealth. Roughly 10% of the sample are in the high-starting-wages/high-wage-growth group. Men are over-represented in the categories with high starting wages, which also consist of individuals with wealthier parents.

For comparison with previous studies, we show the same plot for the mappings $\mathcal{T}_{Levels}$ and $\mathcal{T}_{Fields}$, closely following their definitions in the literature. In panels (a), (c), and (e), the position of all educational programs in the (w_0, g) is identical, but panels (c) and (e) assign those programs to different color-marker groups depending on the level or post-secondary field of education, respectively. $\mathcal{T}_{Levels}$ groups programs based on educational level, i.e., Primary (compulsory schooling), Secondary (high school and vocational degrees), Bachelor (tertiary

⁶We define labor market entry as the year in which individuals complete the highest education obtained before turning 35 if observed or the highest education observed at the oldest age if not turning 35 before 2018.

⁷We focus on the first 9-11 years because wage profiles stabilize later in one's career (Bhuller et al., 2017).

Figure 1: Education-Based Types and their Starting Wages and Wage Growth



Notes: w_0 stands for *average starting wage* and g for *average wage growth*, defined in Section 3.2. The horizontal axis corresponds to the standardized w_0 and the vertical to the standardized g . Points in panels (a), (c), (e) locate all educational programs with at least ten graduates in 2018—described in Section 2—along these two dimensions. Panels (b), (d), (f) show 14 large programs based on the number of graduates observed in the sample in 2018. Colors and markers uniquely assign each program to an education type, depending on the panel's definition.

degrees of four years or less), and Master & PhD (tertiary degrees of five years or more). To construct $\mathcal{T}_{Fields}$ without losing non-tertiary programs, we keep the primary and secondary categories from $\mathcal{T}_{Levels}$ but subdivide the tertiary category based on five post-secondary fields: Humanities, Health & Welfare, Social Science, STEM, and Business.

Types constructed based on the level or field of education overlap substantially in the two

wage outcomes (w_0, g) , and therefore mask important heterogeneity in labor market returns. For example, in Panel (c) for level, starting wages are on average low for Primary (blue squares) and high for Master & PhD (gray diamonds), but many Secondary (red circles) and Bachelor (small orange diamonds) programs have higher starting wages than many Master & PhD programs, and wage growth within each type is widely dispersed. In Panel (e) for fields of study, graduates from STEM (blue diamonds), Business (lilac small triangles), and Social Science (red squares) programs are relatively similar in terms of average wage growth but the variation in starting wages within each group is vast.

To obtain a sense of which specific educational programs are included in the respective clusters, panels (b), (d), and (f) locate 14 large programs in the (w_0, g) plane for the ambition, levels, and fields mappings, respectively. As an example, consider graduates from 5-year business and architecture programs, which face different labor market returns—architects exhibit lower wage growth than business graduates—and are therefore assigned different ambition types. However, they are in one group according to the level of education.

Online Appendix Table A.2 complements this example with a cross-tabulation of ambition types and levels or fields. 10% of graduates who can expect high starting wages and growth do not have a tertiary but a secondary degree. At the same time, more than 40% of individuals in the high-starting-wages/low-wage-growth group have a tertiary degree. This clearly shows that graduating from a university does not guarantee high wage growth. Moreover, Social Science, Business, and STEM fields dominate the group with high starting wages and growth. However, graduates with degrees from the Humanities or Health & Welfare are most commonly found in the groups with low wage growth and either high or low starting wages.

3.3 Alternative Measures of Human Capital

If families value education due to returns on human capital, it is natural to consider two alternative ways of capturing the human capital content of educational programs: the lifetime earnings and labor supply patterns of graduates. First, we use a one-dimensional k-means clustering and identify four groups of programs from low lifetime earnings to high lifetime earnings. Second, we use the LFS and characterize a subset of educational programs based on the fractions of graduates who work part-time or who report working irregular hours (evening, weekend, overtime). In Appendix A.4, we show that there is some overlap between these two categorizations and ambition types, but they capture returns to human capital less precisely. For example, programs whose graduates report working long and irregular hours do not necessarily come

with high wage growth. Some low-wage-growth programs (e.g., teachers, architects) also fall into this category.

3.4 Education-Based Types and Career and Family Outcomes

Finally, we analyze how the different definitions of education relate to career and family outcomes. In Appendix A.5, we first construct seven proxies for career and family outcomes of graduates by educational program, including the wage penalty for part-time work, the fraction of graduates who reach a managerial position, and average accumulated wealth. We use regression analysis to show that the two building blocks of our ambition types—the wage measures w_0 and g , which capture returns to human capital in the labor market—are significantly correlated with these career and family outcomes. Importantly, most coefficients on wage moments remain significant even after controlling for education level, field of study, life-time earnings, and work hours (see Table A.3). We conclude that wage dynamics can explain long-term career and family outcomes even conditional on these widely-used human capital measures.

4 Trends in Assortative Matching

We show that the degree of assortative matching on ambition has increased over time, while assortative matching on educational levels or fields has remained constant since the mid-1990s.

Empirically, the degree of positive assortative matching (PAM) manifests itself as a positive association of spousal types in the cross-section, conditional on marriage. This association can be measured based on correlation coefficients, distance measures, or the frequency distribution of spousal types among couples. However, determining whether PAM has *changed over time* has proven elusive because the educational composition of households evolves for three reasons: (i) changes in the educational attainment of men and women (i.e., the gender-specific *marginal distributions* of education); (ii) changes in singlehood and marriage by gender and education (i.e., the extensive margin decision, *who marries*); and (iii) changes in the joint distribution of husbands' and wives' education (i.e., the intensive margin, *who marries whom*).

Indeed, in Appendix B.1, we show heterogeneous trends in these three margins for all educational categorizations. For example, we observe an increasing number of highly-ambitious individuals, particularly women, which could mechanically increase the probability that two high-type spouses meet and form a couple. Therefore, a more refined measurement of assortative matching is necessary. We employ a measure that is based both on the shares of same-type

couples and the marginal type distributions conditional on marriage: the likelihood ratio (Eika et al., 2019; Chiappori et al., 2025). This measure compares the observed probability that a man of a given type is married to a woman of the same type to that probability under random matching, which is captured by the product of the type-specific shares of men and women in the married population.

Let $P^M(t_m, t_f)$ denote the fraction of married couples of type (t_m, t_f) with male type t_m and female type t_f . Let the fraction of married men of type t_m be denoted by $P^M(t_m)$ ($P^M(t_f)$ is the analogous object for married women). For each couple type (j, j') with $t_m = j$ and $t_f = j'$, the likelihood ratio is defined as follows:

$$s(j, j') = \frac{P^M(t_m = j, t_f = j')}{P^M(t_m = j) P^M(t_f = j')}, \quad (1)$$

where the one-dimensional types $t \in \{1, \dots, j, \dots, T\}$ are the categories defined in Section 3.1: educational level, field, or ambition. The measure relates the observed frequency of a couple type (the numerator) to the expected frequency under random matching (the denominator). A likelihood ratio greater than one implies that more couples form than predicted by random matching. For same-type couples, $(j = j')$, a ratio above one indicates PAM.

We then construct aggregate measure $\mathcal{S}$ as the weighted sum across the type-specific likelihood ratios of same-type couples,

$$\mathcal{S} = s(1, 1) \times \pi_1 + s(2, 2) \times \pi_2 + \dots + s(T, T) \times \pi_T \quad (2)$$

with weights $\{\pi_1, \dots, \pi_j, \dots, \pi_T\}$, $\sum_{j=1}^T \pi_j = 1$, and T is the total number of categories. Chiappori et al. (2020b) suggest to compute the weights as a convex combination⁸ of the male and female married-population shares for type j :

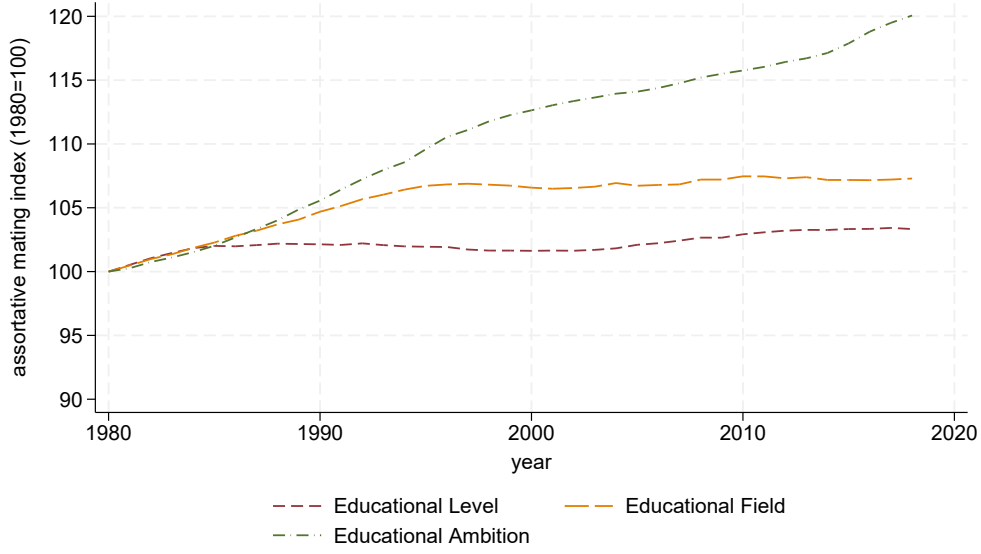
$$\pi_j = \lambda P^M(t_m = j) + (1 - \lambda) P^M(t_f = j). \quad (3)$$

We implement the method by Almar and Schulz (2024) to compute the value of λ that minimizes mechanical distortions caused by changing type distributions, and we use the modal value for λ across years for comparability (see Appendix A.6 for details).

Figure 2 shows the aggregate trends in assortative matching for educational levels (red dashed), fields (orange long dashed), and ambition (green dash-dotted). To facilitate compari-

⁸The weighted sum of likelihood ratios using this weighting scheme fulfills the formal criteria discussed in Chiappori et al. (2020b).

Figure 2: Increasing Assortative Matching based on Educational Ambition



Notes: Shows the assortative matching measure $\mathcal{S}$ (see Equation (2)) for educational levels (red short-dashed line), educational field (orange long-dashed line), and educational ambition (green dash-dotted line). Normalized such that $\mathcal{S}$ in 1980 is 100. Types are constructed as explained in Sections 3.1 and 3.2.

son, we index all trends such that $1980 = 100$.⁹

We find PAM and a strong increase in assortative matching based on educational ambition. Relative to random matching, the likelihood of observing couples with the same ambition type has increased by approximately 20%. In contrast, based on educational levels, we find that PAM has increased by around 3% since 1980, consistent with the flat trend for the US documented by Eika et al. (2019). Based on educational-field types, we find that PAM has increased a bit more (around 7% since 1980), but the trend is flat from the mid-1990s onward.

The difference in trend based on educational levels/fields on the one hand and educational ambition on the other is explained by the differential evolution of the underlying likelihood ratios (1), plotted for all categorizations in Figure B.5 in their unweighted (panels (a), (c), and (d)) and weighted (panels (b), (d), and (e)) versions.¹⁰

For educational levels, the “secondary” category shows an unweighted likelihood ratio that is just slightly above one and flat, indicating that PAM among individuals with secondary education is neither pronounced nor increasing over time. Moreover, the weighted likelihood ratios reveal that the flat trend of the “secondary” group dominates the aggregate trend for educational levels. The tendency to sort decreases in the growing tertiary categories and increases

⁹Because marginal distributions affect $\mathcal{S}$, its levels are not comparable across markets or categorizations. For example, two markets with different education shares but only same-education couples have the same degree of PAM but different $\mathcal{S}$. Yet, our weighting strategy makes the trend within categorization interpretable. Appendix Figure B.4 shows non-indexed trends in $\mathcal{S}$, indicating PAM for all three categorizations ($\mathcal{S} > 1$).

¹⁰Note that the weighted trends add up to the aggregate trends in levels displayed in Figure B.4.

in the shrinking primary education category, but these trends cancel each other out, so there is almost no change in $\mathcal{S}$ for levels.

Any difference between trends for levels and fields in Figure 2 comes from tertiary fields because the primary and secondary categories overlap with the levels categorization. The type-specific likelihood ratios are positive and decreasing in all tertiary fields, similar to the trends for “Master & PhD” (educational levels). For men, the composition of graduates across fields is relatively stable (Figure B.1, Panel (e)). For women, “Business”, “STEM”, and “Social Sciences” expand (Figure B.1, Panel (f)), and their weighted likelihood ratios reveal that they contribute to increasing PAM. However, “Health” and “Humanities” dominate in size, and the weighted measures for these fields decrease after 2000, flattening the aggregate trend.

For educational ambition, the unweighted likelihood measures suggest that PAM is pronounced in the category with high starting wages and high wage growth. Although the unweighted likelihood ratio is decreasing over time, the increasing size of this group, especially for women (Figure B.1, panels (c) and (d)), means that the likelihood of a match among highly ambitious individuals becomes more and more important for the aggregate trend, as revealed by the weighted ratios. We find a stable second-highest likelihood of a match in the category with low starting wages and low wage growth. This group also increases in size and, therefore, contributes substantially to the positive aggregate trend. This is in stark contrast to the shrinking group of individuals with lowest educational level (compulsory schooling). The low-low ambition types include occupations like preschool teachers and physical therapists with secondary or tertiary levels of education, which are growing due to increasing labor demand from the public sector. The other two educational ambition categories exhibit unweighted likelihood ratios close to one. The category with low starting wages but high wage growth shrinks over time, curtailing its influence on the aggregate measure. The category with high starting wages and low wage growth plays the smallest role for the aggregate trend. Taken together, these developments explain the 20% increase in PAM based on educational ambition.¹¹

¹¹As a robustness check, we also consider (log) odds ratios. The odds ratio has some desirable properties (Chiappori et al., 2025) but is local in nature, so interpretation is more involved relative to the aggregate measure $\mathcal{S}$. We calculate it for all the (2×2) submatrices with same-type couples on the diagonal, following Table 6 of Chiappori et al. (2020a). For education levels, all odds ratios decreased between 1980 and 2018. For ambition, we see an even split of increasing and decreasing odds ratios. Consistent with the documented increase in the weighted sum of likelihood ratios, we find that the submatrices with increasing (decreasing) odds ratios cover a larger (smaller) share of the 2018 marriage market.

5 The Marriage Market and Inequality

In this section, we show that the evolution of the composition of households by education affects inequality, but conclusions about the magnitude of this effect depend greatly on the categorization (levels, fields, or ambition). To study the link between the evolution of the composition of households and income inequality for different categorizations, we first use the semi-parametric decomposition technique of [DiNardo et al. \(1996\)](#) (DFL) to quantify how the composition affects inequality along all margins. Second, we combine DFL with the marriage market matching model of [Choo and Siow \(2006\)](#). This allows us to assess the contribution of different margins of marriage market matching (marginal distributions and marital surplus) to raising income inequality between households.¹² Our decomposition builds on [Chiappori et al. \(2020a\)](#), who use a structural model to link marital sorting and inequality. This approach allows us to incorporate both married couples and singles into the analysis.

In Subsection 5.1, we apply the DFL approach to quantify the importance of the changing composition of households. In a nutshell, scenario (i) computes counterfactual income inequality at any year τ by keeping fixed at a baseline year the composition of households along all margins (marginals, who marries, and who marries whom). This approach guarantees that at any year τ the degree of assortative matching $\mathcal{S}$ is as in the base year, which helps us to link trends in assortative matching to trends in inequality.

In Subsection 5.2, we use a structural marriage market matching model of the Separable Extreme Value (SEV) type ([Choo and Siow, 2006](#)) to differentiate between (ii) changing gains from marriage (marital surplus), which affect both the intensive and extensive margin, and (iii) changing marginal type distributions.¹³

5.1 Fixed Household Composition

Let $\mu_{\tilde{x}}$ represent the marital frequency matrix of all types of households—married and single—according to the education categorization resulting from mapping $\mathcal{T}_{\tilde{x}}$, based on characteristics $\tilde{x}$. Note that $\mu_{\tilde{x}}$ is determined by the marginal distributions of types, who marries (the exten-

¹²An alternative way to generate counterfactual household compositions is an algorithmic approach as suggested by [Mosteller \(1968\)](#), and, more recently, [Greenwood et al. \(2014\)](#) and [Eika et al. \(2019\)](#). Our qualitative results are robust to using the algorithmic approach, see an earlier version of this paper ([Almar et al., 2023](#)).

¹³Note that we require additional structure on how individuals choose partners because we cannot fix one margin and determine the other two. This is due to feasibility conditions in the marriage market, which only restrict the allocations to be such that the total mass of individuals of any education type equals the sum of married and single individuals of that type. For example, if we fix the marginal distributions of education at the 1980 level, a decision model is needed to assign individuals into counterfactual fractions of couples and singles by educational type.

sive margin), and who marries whom conditional on marriage (the intensive margin). To reduce notation, in what follows we omit the subscript $\tilde{x}$ when clear from context. Let $\tau \in [1980, 2018]$ denote year, and τ_y and τ_μ denote the year in which household income y and the marital frequency table μ are measured, respectively. Let the distribution of household income conditional on household education types in year τ_y be denoted by $F_{Y|\mu}(y|(t_m, t_f), \tau_y)$.

To analyze how income inequality would have developed had the composition of households remained unchanged, we keep μ fixed at base year τ_μ , while letting the household income distribution F_Y change to some year τ_y . The counterfactual income distribution when income is measured in year τ_y and the household composition is fixed at year τ_μ is:

$$\widehat{F}_Y(y|\tau_y, \tau_\mu) = \int F_{Y|\mu}(y|(t_m, t_f), \tau_y) \psi_{\tau_y, \tau_\mu} d\mu((t_m, t_f)|\tau_y). \quad (4)$$

Importantly, the factor ψ_{τ_y, τ_μ} allows us to re-weight households in the τ_y data to reflect their frequency had the marriage market stayed at its τ_μ composition. Specifically,

$$\psi_{\tau_y, \tau_\mu} = \frac{d\tilde{\mu}((t_m, t_f)|\tau_\mu)}{d\mu((t_m, t_f)|\tau_y)}.$$

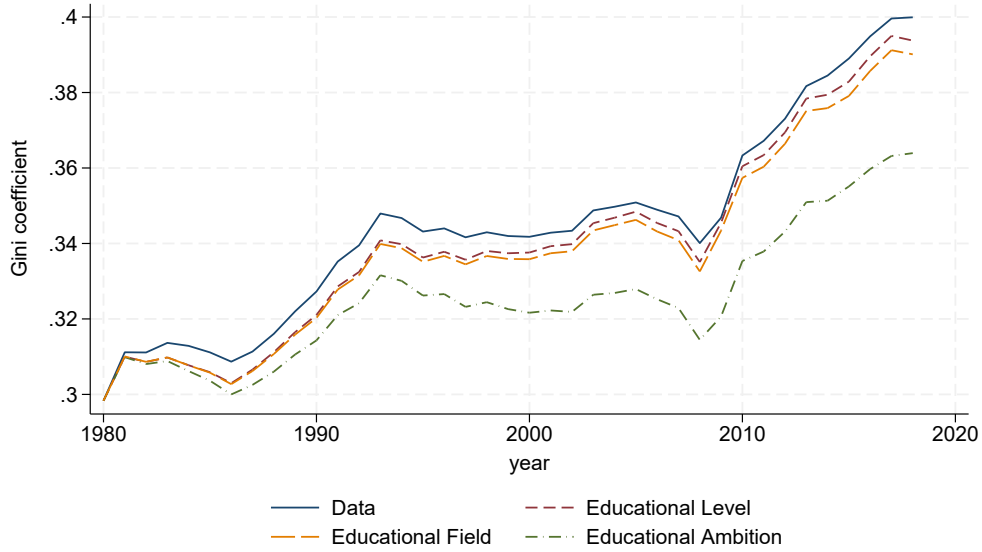
The denominator of the re-weighting factor reflects the composition of households in the observed year τ_y . In practice, we estimate this denominator as the probability of observing household type (t_m, t_f) in year τ_y , which we denote by $P_\mu((t_m, t_f)|\tau_y)$. The numerator, in turn, reflects the *counterfactual* composition of households had the marriage market stayed at its τ_μ composition. We follow DiNardo et al. (1996) and Fortin et al. (2011) and estimate the numerator using the probability of observing household type (t_m, t_f) in year τ_μ , which we denote by $P_\mu((t_m, t_f)|\tau_\mu)$. Hence, in equation (4) we substitute the estimated re-weighting factor

$$\widehat{\psi}_{\tau_y, \tau_\mu} = \frac{P_\mu((t_m, t_f)|\tau_\mu)}{P_\mu((t_m, t_f)|\tau_y)}. \quad (5)$$

for ψ_{τ_y, τ_μ} . Intuitively, household types that are more (less) common in τ_y than in τ_μ get a weight lower (greater) than one in the counterfactual income distribution. We compute the counterfactual income expression (4) and re-weighting factor (5) when $\tau_y = \tau \in [1980, 2018]$ and $\tau_\mu = 1980$: $\widehat{F}_Y(y|\tau_y = \tau, \tau_\mu = 1980)$ and $\widehat{\psi}_{\tau_y = \tau, \tau_\mu = 1980}$. In this counterfactual scenario, the configuration of households stays at its 1980 level for any year τ in which we measure income. Importantly, our choice of weights guarantees that those who are married in year τ exhibit the degree of assortative matching $\mathcal{S}$ of the 1980 households.¹⁴

¹⁴In Appendix A.7, we show this result formally. Additionally, our choice of weights implies that the log odds

Figure 3: Changes in Household Composition and Inequality



Notes: Shows the development of the Gini coefficient of between-household income between 1980–2018 in the data (blue solid line), and in the counterfactual scenario (i) (fixed household composition, explained in Section 5.1) for educational level (red short-dashed line), educational field (orange long-dashed line), and educational ambition (green dash-dotted line) types.

Using counterfactual household income for all education-based categorizations of types, we compute counterfactual moments of income inequality. In Figure 3, we show that if the household composition had stayed fixed as in $\tau_\mu = 1980$, household income inequality at any year $\tau > 1980$ would have been mitigated relative to the trend in the data. We plot, for all years $\tau_y = \tau \in [1980, 2018]$, the observed Gini (blue solid line) and the counterfactual Gini coefficients from scenario (i), holding household composition fixed by educational level (red short-dashed line), educational field (yellow long-dashed), or educational ambition (green dot-dashed). All counterfactual trends lie below the data, indicating that inequality would have increased less. Crucially, however, the counterfactual trend for educational ambition diverges most from the actual trend. Thus, changes in the composition of households based on educational ambition account for a larger share of increasing between-household inequality than the composition of households based on educational level or field. One reason is that highly ambitious women have increasingly entered marriage (the extensive margin), see Figure B.3. Another reason is that positive assortative matching based on educational ambition (the intensive margin) has become more pronounced, while it has barely increased based on levels and fields, recall Figure 2.

In Table 1, we decompose the total change in between-household income inequality between 1980 and 2018. Column (a) shows the results for the Gini coefficient, which summarizes in-
ratios are also held fixed at their 1980 values.

equality in the entire distribution; in columns (b) and (c) we extend our analysis and show the 90/50 and 50/10 percentile ratios, respectively. For each inequality measure, the first row contains the inequality changes in the data (Δ_{Data}). Between-household income inequality has increased according to all three measures. The Gini coefficient has increased by 0.102 (from 0.298 to 0.400), the 90/50 percentile ratio has increased by 0.322 (from 1.643 to 1.957), and the 50/10 percentile ratio has even increased by 3.626 (from 3.170 to 6.796). These changes correspond to 100%.

Panel (i) of Table 1 shows our main result: with a fixed household composition in terms of educational ambition types, the counterfactual 2018 Gini coefficient amounts to 65% of the true value. That is, 35% of the increase in the Gini since 1980 can be explained by changing household composition based on educational ambition. Using the common education-based categorizations, we do not arrive at the same conclusion. With 94% for levels and 90% for fields, the household composition matters much less for the observed inequality increment according to these categorizations.

The composition of the households based on educational ambition also stands out by explaining a greater share of increasing inequality for percentile-based measures. The increase in the 90/50 ratio is close to the factual increase based on fields and levels, but based on educational ambition it would only have increased by 50%. For the 50/10 ratio, the increase is even mitigated by the changing composition of households through the lens of levels and fields, while the compositional changes in terms of ambition amplified household income inequality in the lower half of the income distribution.

Overall, these differences highlight our main contribution: how we define education-based types matters for conclusions on how the marriage market affects inequality between households.

5.2 Fixed Marital Surplus vs. Fixed Marginal Distributions

We investigate what particular aspect of the marriage market contributed to amplifying the increase in household income inequality. Because the supply of education-based types, the decision to marry, and the decision of whom to marry are endogenously related, disentangling their contribution requires additional structure. We follow the literature (Choo and Siow, 2006; Dupuy and Weber, 2022; Chiappori et al., 2020a,b, 2025) and use the Separable Extreme Value (SEV) model.

Specifically, we decompose the role of trends in household composition to explain changes in inequality into the contributions of changing marginal distributions of education and the

Table 1: Decomposing Changes in Income Inequality

	(a) Gini		(b) P_{90}/P_{50}		(c) P_{50}/P_{10}	
Factual change (Δ_{Data})	0.102	100%	0.322	100%	3.626	100%
	Δ_{Gini}	$\frac{\Delta_{Gini}}{\Delta_{Data}}$	$\Delta_{P_{90}/P_{50}}$	$\frac{\Delta_{P_{90}/P_{50}}}{\Delta_{Data}}$	$\Delta_{P_{50}/P_{10}}$	$\frac{\Delta_{P_{50}/P_{10}}}{\Delta_{Data}}$
(i) Fixed household composition						
Educational Level	0.095	94%	0.318	99%	4.091	113%
Educational Field	0.092	90%	0.301	93%	4.063	112%
Educational Ambition	0.066	65%	0.161	50%	2.531	70%
(ii) Fixed marital surplus						
Educational Level	0.078	76%	0.225	70%	2.417	67%
Educational Field	0.078	77%	0.221	69%	2.538	70%
Educational Ambition	0.076	75%	0.206	64%	2.547	70%
(iii) Fixed marginals						
Educational Level	0.134	132%	0.485	150%	6.953	192%
Educational Field	0.132	130%	0.471	146%	6.965	192%
Educational Ambition	0.105	104%	0.319	99%	4.808	133%
(iii.a) Fixed marginals (male)						
Educational Level	0.103	102%	0.313	97%	4.401	121%
Educational Field	0.102	100%	0.309	96%	4.419	122%
Educational Ambition	0.099	97%	0.301	93%	3.889	107%
(iii.b) Fixed marginals (female)						
Educational Level	0.129	127%	0.426	132%	5.873	162%
Educational Field	0.129	127%	0.426	132%	5.898	163%
Educational Ambition	0.108	107%	0.335	104%	4.457	123%

Notes: Panels show the counterfactual scenarios constructed as explained in Section 5.1 (panel (i)) and in Section 5.2 (panels (ii) to (iii.b)). P_{90}/P_{50} (P_{50}/P_{10}) is the ratio of the 90th and 50th (10th) percentile in the income distribution. Δ_{Data} shows the observed inequality changes. $\Delta_{Gini}/\Delta_{Data}$ is the counterfactual change relative to the observed change. Each row within each scenario shows the columns' statistic for one of the three definitions of types (as explained in Section 3): educational level, educational field, and educational ambition types.

evolving value of marriage relative to singlehood (marital surplus). To do so, we combine the DFL approach—which we use to construct the re-weighting factors—with the SEV structure—which allows us to isolate the role of changing marginal distributions from that of changes in the value of marriage. The framework introduced by Choo and Siow (2006) models individuals' decision of whom to marry or whether to remain single based on the value of their potential matches relative to the value of remaining single: the marital surplus. A distinctive feature is that the marital surplus is the sum of two components: a *systematic* component that only depends on the type composition of the potential couple, and a random idiosyncratic taste shock that depends only on the type of the potential partner.¹⁵ Importantly, Choo and Siow (2006)

¹⁵While the advantage of Choo and Siow (2006) is its simplicity, a number of papers criticize the underlying—admittedly strong—assumptions (see, e.g., Galdani and Sinha, 2023). One such assumption is independence-of-irrelevant-alternatives (IIA). Gutierrez (2020) relaxes IIA by modifying the Choo and Siow (2006) model and considering match-specific rather than type-specific taste shocks. We have implemented this modification and

derive a closed form relationship between the systematic component of the marital surplus of couple type (j, j') —which we denote by $\Pi_{jj'}$ —and the ratio of the observed share of couples type (j, j') and the geometric average of the single shares of type j and j' :

$$\Pi_{jj'} = \frac{P_\mu(t_m = j, t_f = j')}{\sqrt{P_\mu(t_m = j, t_f = \emptyset) P_\mu(t_m = \emptyset, t_f = j')}}, \quad (6)$$

where $P_\mu(t_m = j, t_f = \emptyset)$ ($P_\mu(t_m = \emptyset, t_f = j')$) is the probability of observing a man of type j (women of type j') without a partner.

We combine the marital surplus (6) with the feasibility conditions in the marriage market whereby the mass of individuals of type t equals the sum of the mass of individuals of that type who marry and remain single. We denote the mass of single women type t_f and of single men type t_m by $P_\mu(\emptyset, t_f)$ and $P_\mu(t_m, \emptyset)$, respectively. For a particular categorization based on $\tilde{x}$, we denote the mass of women type t_f and the mass of men type t_m , respectively, by

$$\begin{aligned} P_{\tilde{x}}(t_f) &= \sum_{t_m} P_\mu(t_m, t_f) + P_\mu(\emptyset, t_f) \\ P_{\tilde{x}}(t_m) &= \sum_{t_f} P_\mu(t_m, t_f) + P_\mu(t_m, \emptyset). \end{aligned} \quad (7)$$

Formally, let $\tau_{\tilde{x}}$ be the year in which the marginal distribution of types based on definition $\tilde{x}$ is measured, and let τ_Π be the year in which the marital surplus is measured. We define the income distribution when household income is measured in year τ_y , the marginal distributions of $\tilde{x}$ in $\tau_{\tilde{x}}$, and the marital surplus Π in τ_Π as

$$\widehat{F}_Y(y|\tau_y, \tau_{\tilde{x}}, \tau_\Pi) = \int F_{Y|\mu}(y|(t_m, t_f), \tau_y) \psi_{\tau_y, \tau_{\tilde{x}}, \tau_\Pi} d\mu((t_m, t_f)|\tau_y). \quad (8)$$

As in counterfactual (i) the factor $\psi_{\tau_y, \tau_{\tilde{x}}, \tau_\Pi}$ allows us to re-weight households in the τ_y data to reflect their frequency had the education marginals and marital surplus stayed at their $\tau_{\tilde{x}}$ and τ_Π compositions, respectively. In this case, in which we distinguish between marginals and surplus, we have

$$\psi_{\tau_y, \tau_{\tilde{x}}, \tau_\Pi} = \frac{d\widehat{\mu}((t_m, t_f)|\tau_{\tilde{x}}, \tau_\Pi)}{d\mu((t_m, t_f)|\tau_y)}.$$

Again, the numerator of the re-weighting factor is not observed, so it has to be estimated. An additional complication is that when $\tau_{\tilde{x}} \neq \tau_\Pi$, we do not have a frequency table $P_\mu((t_m, t_f)|\tau_{\tilde{x}}, \tau_\Pi, \tau_{\tilde{x}} \neq \tau_\Pi)$ as a data counterpart, because only marriage markets in which the marginal type distri-

present results for the model-based counterfactuals in Appendix Table B.2. All our conclusions are unaffected but quantitatively the role of the marriage market is slightly reduced in all scenarios.

bution and the marital surplus are measured in the same year ($\tau_{\tilde{x}} = \tau_{\Pi}$) realize. We retrieve these counterfactual frequency tables, $\hat{\mu}$, using the model. For example, when we fix marginals (7) at time $\tau_{\tilde{x}}$ and let the marital surplus be as observed in year $\tau > \tau_{\tilde{x}}$, we use the system of equations (6) and (7) in two steps: first, we compute the *counterfactual* fractions of singles that would close the marriage market in τ when the marginal distributions are as in $\tau_{\tilde{x}}$; second, we compute the implied *counterfactual* fractions of couples. We end up with a counterfactual matching matrix, $\hat{\mu}$, specifying the frequency of each type of household—including the singles—with elements denoted by $P_{\hat{\mu}}^{SEV}((t_m, t_f)|\tau_{\tilde{x}}, \tau_{\Pi})$. Once we have calculated the counterfactual household compositions, $\hat{\mu}$, in which the marginals are measured in year $\tau_{\tilde{x}}$ and the surplus in year τ_{Π} , we can estimate the re-weighting factor by

$$\hat{\psi}_{\tau_y, \tau_{\tilde{x}}, \tau_{\Pi}}^{SEV} = \frac{P_{\hat{\mu}}^{SEV}((t_m, t_f)|\tau_{\tilde{x}}, \tau_{\Pi})}{P_{\mu}((t_m, t_f)|\tau_y)}. \quad (9)$$

and use it to compute the different counterfactual income distributions in (8).

In the fixed marital surplus counterfactual (ii), we fix marital surplus at $\tau_{\Pi} = 1980$, while we measure income and education marginals in any year $\tau > 1980$. Specifically, we use counterfactual re-weighting factor $\hat{\psi}_{\tau_y=\tau, \tau_{\tilde{x}}=\tau, \tau_{\Pi}=1980}^{SEV}$ to obtain the counterfactual income distribution $\widehat{F}_Y(y|\tau_y = \tau, \tau_{\tilde{x}} = \tau, \tau_{\Pi} = 1980)$. Intuitively, every year $\tau > 1980$ each household—coupled or single-headed—is reweighted so that the relative frequencies of couples to singles by type and gender give rise to the gains of marriage relative to singlehood revealed in the year 1980.

In the fixed-marginals counterfactual (iii), we fix the marginal distributions of education at $\tau_{\tilde{x}} = 1980$, while we measure income and marital surplus at any year $\tau > 1980$. Specifically, we use counterfactual re-weighting factor $\hat{\psi}_{\tau_y=\tau, \tau_{\tilde{x}}=1980, \tau_{\Pi}=\tau}^{SEV}$ and counterfactual income distribution $\widehat{F}_Y(y|\tau_y = \tau, \tau_{\tilde{x}} = 1980, \tau_{\Pi} = \tau)$. We also keep either the male (case iii.a) or the female (case iii.b) marginal type distributions fixed. In those cases, we further distinguish between marginals by gender in expressions (8) and (9).

Panel (ii) of Table 1 shows the results for the fixed-surplus scenario. When we generate households in 2018 based on the 1980 gains from marriage and the 2018 marginal distributions, we find that inequality would have increased about a quarter less than in the data. Interestingly, this result is quantitatively similar across categorizations, so differences in their evolution of marital surplus are not driving the result in scenario (i). We reach similar conclusions when we measure income inequality based on percentile ratios.

In Panel (iii) of Table 1, we first keep the marginal type distributions for both genders fixed at the 1980 level. We then repeat the exercise keeping either the male or the female marginal

type distribution fixed, see Panels (iii.a) and (iii.b). The marginal distributions shifted such that the numbers of individuals in the top categories increased, and this change is more pronounced for women.¹⁶ That is, there are more men and women who graduate with tertiary degrees and/or from ambitious educational programs in 2018 compared to 1980.

Based on the Gini coefficient in column (a) and educational levels (or fields), we find that these trends in educational attainment had a mitigating effect on inequality. Without the shift, inequality would have increased to 132% (130%) of the actual 2018 value. The mitigating effect manifests itself mainly in the lower half of the income distribution. The 50/10 percentile ratio in column (c) would have been almost twice as high. For the 90/50 ratio in column (b), we see slightly more modest mitigating effects for educational level and field types (150% and 146%). Based on educational ambition types, we do not find significant mitigation (or amplification) for the Gini or the 90-50 ratio with fixed marginals (104% and 99%), but there is a mitigation effect in the lower half of the distribution (133%).

Fixing only women’s marginals at 1980 in scenario (iii.b) essentially reproduces scenario (iii) with both marginal distributions fixed across categorizations. The mitigating effects remain but become slightly weaker for level and field. By contrast, fixing only male marginals (iii.a) implies little mitigation: the counterfactual distributions nearly match the data, aside from a modest reduction in the 50/10 ratio for level and field. Thus, the mitigation is driven primarily by changes in female marginals.

The counterfactuals with fixed marginals for women highlight again the importance of distinguishing between the three categorizations: For example, while women’s move into tertiary education had a considerable mitigating effect, their entry into (ambitious) high-wage/high-growth programs did not mitigate the rise in inequality. This, in turn, contributes to a larger overall role of the marriage market for inequality trends based on ambition types.

6 Alternative Categorizations and Robustness

We explore the meaning of our results further in several ways. First, we replicate our main analysis using two alternative categorizations based on lifetime earnings and labor supply moments in Appendix B.3. Trends in assortative matching on hours moments and lifetime earnings are flatter than on ambition. This suggests that the increasing spousal similarity in ambition is due to sorting on career trajectories that lifetime-income and hours-based classifications are unable

¹⁶The share of men (women) with long-cycle tertiary education increased by a factor of 3 (11) between 1980 and 2018. For educational ambition types, the share of men (women) in the top category doubled (increased five-fold), as shown in Figure B.1 in the Online Appendix.

to distinguish. However, changes in the composition of households based on these alternative categorizations have affected inequality more than what educational levels or fields suggest. Consistent with the results for ambition, we find that inequality growth would have been mitigated had the composition of households in terms of work hours or lifetime earnings remained unchanged. However, for these categorizations, changes in the composition of *different-type* couples have had the greatest influence on inequality.

Second, in Appendix B.5 we investigate whether our main results are robust to constructing ambition types in alternative ways. We construct ambition types by gender and by cohort, consider different numbers of clusters, use part-time penalties (Goldin, 2014) and manager premia as alternative clustering variables, and apply our approach to more aggregated data by level-and-subfield. Our conclusions do not change as long as the variables used to group units capture returns to human capital.

Third, our results remain when we exclude individuals with coarser 3-digit educational codes—commonly found for immigrants who completed studies abroad—and when we restrict the sample to individuals aged 40 to 49—who have most stable family and career circumstances.

Finally, we find a similarly important role of the marriage market for inequality based on ambition types when measuring inequality trends based on disposable income. This indicates that the main results are not substantially mitigated by taxation and transfers.

7 Conclusion

We provide new insights into the relationship between education-based marital sorting and between-household inequality. Our main finding is that conclusions about the link between household composition and inequality greatly depend on how education-based types are constructed.

We build a novel categorization of education by clustering education programs based on the wage dynamics (starting wages, early-career growth) of graduates and label the resulting categorization *educational ambition*. Going beyond alternative classifications based on educational level, field of study, labor supply patterns, or lifetime income, these ambition types are constructed to reflect differences in the labor market returns to human capital across educational programs, and they are strong predictors of family and career outcomes.

Positive assortative matching on educational ambition increased by about 20% between 1980 and 2018, while PAM on the alternative classifications remained relatively flat.

Overall, changes in the composition of households by educational ambition explain approx-

imately 35% of the increase in between-household inequality over the same period.

Taken together, our analysis suggests that considering richer classifications than the level or field of education is a promising direction for future research on the relationship between marriage and labor markets. Various administrative data sources across countries provide long panel data with program-level information that allows researchers to implement our approach to define marital types. Furthermore, we find similar results when defining educational ambition based on more aggregate units of educational levels by subfield. This suggests that coarser data can also be used to construct ambition types, as long as those data include some information about labor market returns to human capital in the form of wage dynamics or other measures of career trajectories of graduates.

Data Availability

The micro data we use in our paper is owned by Statistics Denmark. We (the authors) can therefore not license users to download, copy and modify the microdata. However, our data access is not exclusive. In the interest of scientific validation of analyses published using Statistics Denmark microdata, the Department of Economics and Business Economics, Aarhus University, will assist researchers in obtaining access to the analysis data set. Access is conditional on a contract with Statistics Denmark that has to be made with Statistics Denmark, accepting their conditions for data access, see <http://www.dst.dk/en/TilSalg/Forskningsservice/Dataadgang.aspx>. If granted, the access is strictly limited to the replication exercise. Access requests can be emailed to: datamanager@econ.au.dk.

The Department of Economics and Business Economics, Aarhus University, guarantees that the final analysis data sets will be archived for at least 5 years. As documented in the paper, the final analysis data sets are constructed from raw register data using SAS. We archive the final analysis data sets along with the SAS code used to generate it on servers hosted by Statistics Denmark. The empirical analyses in the paper were done in Stata. All Stata programs used to process the raw data and generate the results, figures, and tables in the paper are available in our replication package (Almar et al., 2025). The documentation of the replication package includes details and concise instructions on how to run the programs that generate the results in the paper.

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